INCH - POUND

MIL-T-55207C(MI)
1 April 1991
SUPERSEDING
MIL-T-55207B(MI)
8 February 1980

MILITARY SPECIFICATION

TEST SET, ELECTRICAL, CABLE: AN/GSM-45

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a test set for use in testing various cable assemblies having 126 conductors or less by connecting both ends or only one end of the cable through a coupling medium to the test set. The test set also has the capability of testing up to six coaxial conductors including the outer conductor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6625
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-F-14252	-	Fire Control Materiel: General Specification Governing the Design and Inspection of

STANDARDS

MILITARY

MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

A8213077	-	Test Set, Electrical Cable, AN/GSM-45
F10512452	-	Schematic Diagram, Test Set Cable, AN/GSM-45

(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication. The test set shall be manufactured in accordance with the applicable drawing (see 2.1.2) and all drawings specified therein. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence to insure inherent quality and performance capabilities.

3.4 General specifications. The contractor shall be responsible for adherence to and compliance with 3.4.1 and 3.4.2.

3.4.1 Dimension and tolerance. Dimension and tolerance shall be in accordance with MIL-F-13926.

3.4.2 Special requirements. Special requirements shown below shall be in accordance with MIL-F-14252.

a. Service conditions
b. General responsibilities of contractors
c. Parts and materials
d. Use of standard items
e. Finishes
f. Lubrications
g. Wired equipment
h. Identification
i. Workmanship.

3.5 Environmental.

3.5.1 Storage temperature. The test set shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature 16 to 32° Celsius (C) after having been exposed and thermally stabilized at ambient temperatures of 82° C and -62° C (see 4.9.1.1).

3.5.2 Operating temperatures. The test set shall meet all the performance requirements of this specification while exposed and thermally stabilized at ambient temperatures of 60° C and -40° C (see 4.9.1.2).

3.5.3 Humidity and condensation.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the currentversion before use.

3.5.3.1 Humidity. The test set shall meet all the performance requirements specified herein when thermally stabilized at the temperature and relative humidity (RH) specified below (see 4.9.1.3).

a.	4° C	95% RH
b.	25° C	50% RH
c.	25° C	100% RH
d.	49° C	50% RH
e.	60° C	25% RH

3.5.3.1.1 Condensation. The test set shall meet all the performance requirements of this specification when exposed to four 24-hour humidity condensation cycles in accordance with figure 1 (see 4.9.1.3).

3.5.4 Altitude. The test set shall meet the low temperature performance requirements specified herein at altitudes ranging up to 10,000 feet above sea level after being exposed to an altitude of 50,000 feet for 16 hours (see 4.9.1.4).

3.5.5 Rain. The test set shall conform to procedure I of MIL-F-13926.

3.5.6 Salt spray. The test set shall conform to method 509.3, of MIL-STD-810, except that length of the test shall be 50 hours (see 4.9.1.9).

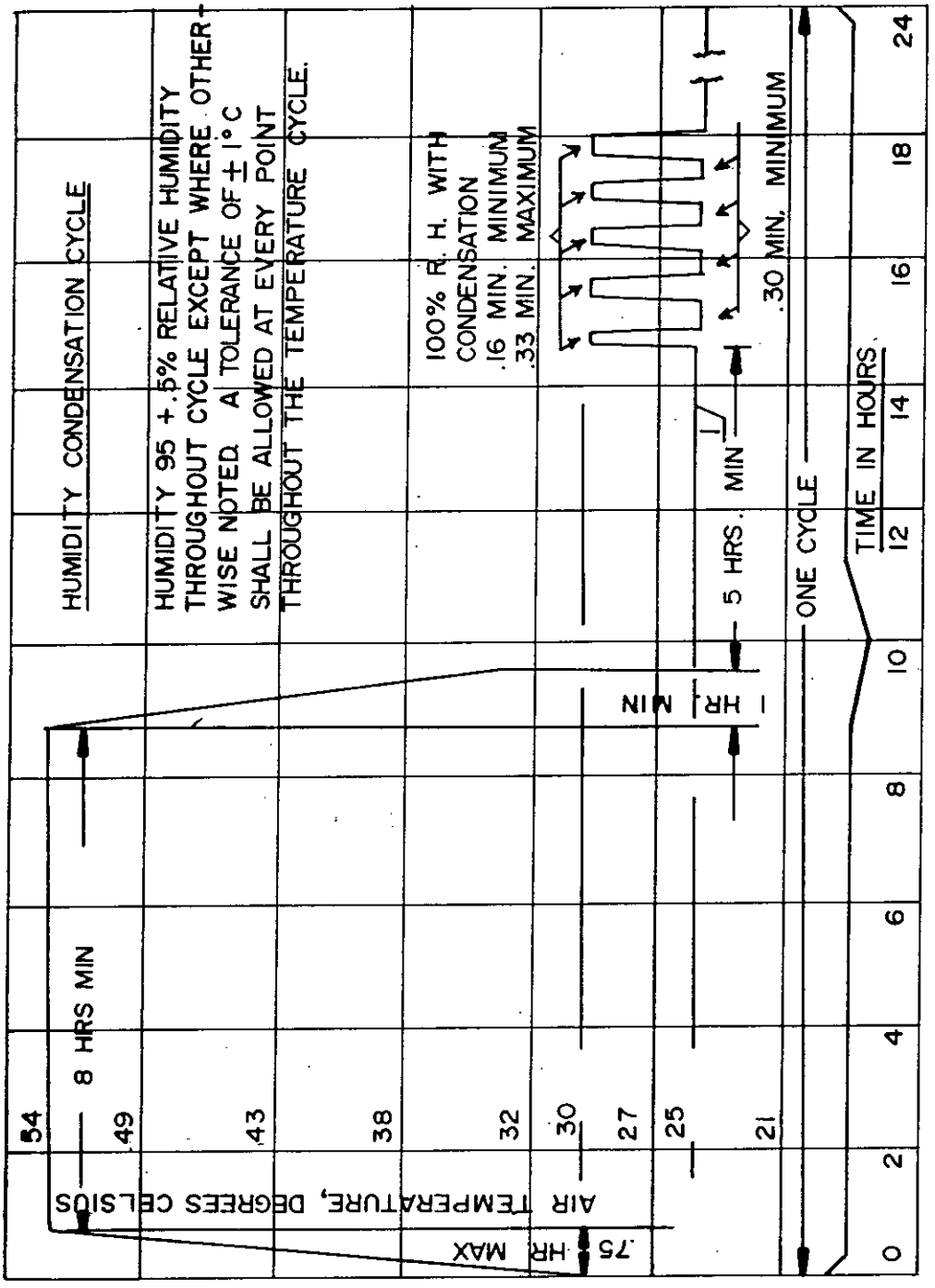
3.5.7 Shock. The test set shall show no evidence of damage or fail to meet the performance requirements specified herein after being subjected to 18 shocks, 6 in each of 3 mutually perpendicular directions, of 30-g acceleration for a duration of 11 milliseconds each (see 4.9.1.5).

3.5.8 Vibration. The test set shall show no evidence of damage or fail to meet the requirements of 3.6.3 when tested in accordance with procedure III of MIL-F-13926, except that the frequency limits, amplitude, acceleration, time, and temperature specified herein shall apply (see table I and 4.9.1.6).

3.6 Electrical.

3.6.1 Power requirements. The test set shall operate at 115 or 230 volts alternating current, $\pm$ 10 percent, at the frequency range of 60 $\pm$ 5 to 400 $\pm$ 10 Hertz (Hz) and shall meet the requirements specified herein (see section 4).

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.



NOTES:

1/ Cycle is as specified or until equipment stabilizes (whichever is longer).

FIGURE 1. Humidity Condensation Cycle.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE I. Vibration test procedures.

Frequency limits	Amplitude	Acceleration	Time	Temperature
5 - 10 HZ	0.030	.....	15 min	-54° C, room + 74° C
11 - 50 HZ	0.015	.....	30 min	-54° C, room + 74° C
51 - 400 HZ	.....	.....	30 min	-54° C, room + 74° C

3.6.2 Cycling time. The test set shall be capable of testing a cable with up to 126 conductors within 140 seconds per test (see 4.9.6) for the following tests:

a. Continuity resistance, single-ended
b. Shorts, single-ended
c. Transposed conductors, double-ended
d. Continuity resistance, double-ended
e. Insulation resistance, 500 V.D.C. test potential
f. Insulation resistance, 1,000 V.D.C. test potential

3.6.3 Test functions.

3.6.3.1 Continuity resistance, single-ended. The test set shall test and indicate, both manually and automatically, conductor continuity resistance with an acceptability level variable from .2 to 5,000 ohms at an accuracy of ± 10 percent with one end of the cable connected to the test set. Single-ended testing shall be accomplished by means of a shorting plug at the opposite end of the cable (see 4.9.7 and 4.9.15.1).

3.6.3.2 Shorts. The test set shall detect and indicate, both manually and automatically, shorts between each conductor against all others connected together and against each conductor to the connector (see 4.9.8).

3.6.3.3 Transposed conductors. The test set shall detect and indicate, both manually and automatically, transposed conductors and shall perform a search function to locate the transposition (see 4.9.9).

3.6.3.4 Continuity resistance, double-ended. The test set shall measure and indicate, both manually and automatically, conductor continuity resistance with an acceptability level variable from .2 to 5,000 ohms and accuracy of ± 10 percent with both ends of the cable connected to the test set (see 4.9.10).

3.6.3.5 Insulation resistance. The test set shall measure and indicate, both manually and automatically, conductor insulation resistance to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

all other conductors connected together and each conductor to the connector with an insulation resistance level variable from 2 to 1,000 megohms and an accuracy of ± 10 percent (see 4.9.11). This test shall be run at high voltage potentials of 500 and 1,000 volts direct current limited to 15 milliamperes. This test shall also be capable of being run double- or single-ended as specified in 3.6.3.2 for shorts (see 4.9.15.2).

3.6.3.6 High voltage adjustment and calibration. The test set shall have a means of adjusting the 500 and 1,000 volt potentials used in insulation resistance tests by front panel controls and meter.

3.6.3.7 Ohmmeter. The test set shall be capable of being used as an ohmmeter from 0 to 1,000 megohms at an accuracy of ± 10 percent with the following ranges (see 4.9.13):

a. R x 1 (Direct reading .2 to 1,000 ohms, 10 ohms midscale. Accuracy shall be ± 10 percent)

b. R x 10
c. R x 100
d. R x 1K
e. R x 10K
f. R x 100K
g. R x 1 megohm.

At each of the above ranges, the test set shall provide adjustment for each scale extreme with its associated control, with not less than a 30° rotational adjustment still available.

3.6.3.8 Radio Frequency (R. F.) attenuation. The test set shall measure the R. F. attenuation of coaxial conductors up to six in number, selected manually. The attenuation shall be measured at a frequency of 100 Megahertz ± 10 percent in a range of 0 to 100 ± 15 decibels (db)/10 db. The substitution method employing a zero indicating meter shall be incorporated in the test set (see 4.9.14 and 4.9.15.3).

3.6.4 Indicators. The test set shall provide a visual indication at all times of the conductor being tested and in the case of the transposed conductor test, a direct numerical readout of the termination of the transposed conductor found in the search shall be made. The test set shall also provide a "Go, No-Go" light and meter readouts as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE II. Go, no-go indications.

Test	Test cable termination	Go and no-go lights	Meter
Continuity resistance	single-ended	X	X
Shorts	single-ended	X	
Transferred conductor	double-ended	X	
Continuity resistance	double-ended	X	X
Insulator resistance 500V	single-ended or double-ended	X	X
Insulator resistance 1000V	single-ended or double-ended	X	X
Ohmmeter			X
Attenuation			X

3.6.5 Operation. The test set shall switch from automatic operation to manual operation and back to automatic operation, on or during any test except a search for transposed conductors, ohmmeter and attenuation.

3.6.6 Common return. The test set shall, in single-ended continuity resistance testing, provide a common return by connecting all conductors except the one under test as the return.

3.6.7 Output voltage. The test set shall be capable of supplying 28 ± 4 V.D.C. at 0.25 amperes from a front panel mounting connector.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection condition. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.6, 4.7, 4.8, and 4.9 in this specification.

4.4 First article inspection. Unless otherwise specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. This inspection shall be conducted at the contractor's facility, or Government approved laboratory using test equipment specified in the contract. The sample shall be inspected for compliance with all requirements specified herein. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.5 Quality conformance inspection. Quality conformance inspection shall include inspections and tests as specified in 4.6 through 4.9.

4.5.1 Lot formation. An inspection lot shall consist of a group of like items produced at the same utilizing the same batches of materials, lots of components, process runs, fabrication techniques, assembly techniques, tools, equipment, and facilities, but shall not exceed one month's production. Upon submittal to the procuring activity for acceptance, each lot shall be accompanied by in-process inspection records and inspection results as required by the procuring activity inspector and this specification.

4.6 Inspection of components. All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of the test set shall be inspected in accordance with the written inspection system required by the contract or order (see 6.2).

4.6.1 Components sampling. Sampling for inspection of components shall be in accordance with table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE III. Sampling plan for components.

Lot Size	Sample Size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.7 Inspection equipment. Supply, calibration, maintenance, and use of inspection equipment shall be in accordance with the contract or order (see 6.2).

4.7.1 Accuracy requirements. Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.7.2 and 4.7.3. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 Inspection equipment design requirements. The accuracy of test equipment used to perform the acceptance tests specified below shall be equal to or greater than the following maximum allowable tolerances on test measurements unless otherwise specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Tolerances and test conditions.

Test condition	Tolerances
Temperature:	Storage and operating ± 2° C. The test facilities shall be such that the equipment under test shall be not greater than 50 percent of the internal volume of the test chamber and no portion shall be closer than 6 inches from any surface of the chamber.
Altitude:	± 5 percent in feet.
Humidity:	± 5 percent relative.
Vibration Amplitude:	± 5 percent. This tolerance applicable only to the frequency measuring instruments.
Vibration Frequency:	± 2 percent. This tolerance applicable only to the frequency measuring instruments.
Shock Test:	Shock intensity shall be within ± 10 percent when measured with a filter having a cutoff to eliminate all harmonics.
Rain Test:	Rain shall be 4 + 1, - 0 inches per hour, measured per hour with a U.S. Weather gage at the surface of the equipment.

4.7.3 Additional test equipment. The following additional items are required to perform the final acceptance inspection of the test set and must conform to the accuracies specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE V. Additional test equipment accuracies.

Item	Accuracy
1. Production test cable	Complete 125 conductor cable, no faults. T2 tied to U-13 for end of test.
2. Decade box containing precision resistors: 10 ohms 1 megohm 100 ohms 2 megohms 1K ohm 100 megohms 100K ohms 200 megohms 100K ohm	0.25 watt ± 1 percent of rated value.
3. R. F. Attenuator	
Range	
1 db at 50 ohm impedance	.15 db/10db
2 db at 50 ohm impedance	.15 db/10db
3 db at 50 ohm impedance	.15 db/10db
6 db at 50 ohm impedance	.15 db/10db
10 db at 50 ohm impedance	.15 db/10db
20 db at 50 ohm impedance (4 each)	.15 db/10db
4. Voltmeters (a) 100-250 V.A.C. 55-410 Hz	These meter readings shall fall within the 30 to 80 percent of scale range. The overall scale range shall be within an accuracy of ± 2 percent.
5. Multimeter 0-1200 V.D.C. ± 2 percent, 20,000 ohm/Volt	± 2 percent at full scale.

4.8 Acceptance inspection.

4.8.1 Classification of defects. The classification of defects listed in table VI shall constitute the minimum inspection required to be performed prior to acceptance or rejection by lots. Sampling shall be in accordance with table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification of defects.

Categories	Defects	Method of inspection	Sampling
101	Incomplete assembly	Visual (VIS)	Table VII
102	Continuity, single-ended cable	Standard Meas. Equip. (SME)	Table VII
103	Continuity, double-ended cable	SME	Table VII
104	Shorts	SME	Table VII
105	Improper cycling time	SME	Table VII
106	Transposed conductors	SME	Table VII
107	Improper insulation resistance	SME	Table VII
108	Operability at high voltage potential	SME	Table VII
109	Improper ohmmeter adjustment	SME	Table VII
110	Inability to adjust and calibrate high voltage	SME & Tactile (TAC)	Table VII
111	Inaccurate R. F. readings	SME	Table VII
112	Improper Indications	SME	Table VII
113	Improper Switching Operations	VIS & TAC	Table VII
114	Improper single-ended Continuity	SME	Table VII
115	Inaccurate output voltage	SME	Table VII

TABLE VII. Sampling plan for acceptance.

Lot Size	Sample Size
2 to 13	All
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

4.9 Acceptance testing.

4.9.1 Environmental testing. Environmental tests specified herein shall be performed on the first article sample and 1 in every 25 thereafter. This sequence of selecting a test set for environmental tests shall apply only after the first article sample has been accepted by the Government representative. The quantities specified above shall prevail providing the contractor continues manufacturing the test set using the same processes and techniques used to produce the production sample. Any alterations in the manufacturing techniques or processes will require a new production sample and the sampling procedure shall begin again. The contractor shall also provide a new production sample when any contract revision, determined by the contracting officer, indicates a change in performance requirements.

4.9.1.1 Temperature test (storage). The test set, exposed with the cover off, shall be subjected to test procedure II of MIL-F-13926 using the temperature specified in 3.5.1, and shall conform to the requirements specified herein.

4.9.1.2 Temperature testing (operating). The test set, exposed with the cover off, shall be subjected to test procedure I of MIL-F-13926 using the temperature specified in 3.5.2 and shall conform to the requirements specified herein.

4.9.1.3 Humidity test. The test set, exposed with the cover off, shall be maintained at the temperature and relative humidity specified in 3.5.3 until thermal stabilization is reached, and shall conform to the requirements of 3.5.3.

4.9.1.4 Altitude test. The test set, exposed with the cover off, shall be maintained at a temperature of -40° C for a sufficient time to obtain thermal stability. The pressure shall then be lowered to simulate an altitude of 10,000 feet. The test set shall then conform to the requirements specified in 3.5.4.

4.9.1.5 Shock test. The test set, with the cover on, shall be subjected to the test specified in procedure II of MIL-F-14252 except the shock shall be 30g. The test set shall conform to the requirements specified in 3.5.7. Time duration shall be 11 milliseconds $\pm$ 1 millisecond.

4.9.1.6 Vibration test. The test set, with the cover on, shall be subjected to the vibration test specified in procedure III of MIL-F-13926, except that frequency amplitude, acceleration and time shall be as in 3.5.8. The test shall be conducted -54° C, at room temperature, and at 71° C. Test sets shall be inspected after each directional change and after each temperature change for conformance with 3.5.8.

4.9.1.7 Humidity and condensation test. The test set, with the cover off, shall be exposed to four 24-hour humidity condensation cycles in

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

accordance with figure 1. The test set shall conform to the requirements of 3.5.3.1.1 after each cycle at the room temperature phase of the cycle after condensation. Temperature, time, and relative humidity tolerances specified in figure 1 shall apply.

4.9.1.8 Rain test. The test set, with the cover on, shall be subjected to rain in accordance with procedure I of MIL-F-13926. The test set shall conform to the requirements specified in 3.5.5.

4.9.1.9 Saltspray test. The test set, with the cover on, shall be subjected to a 50-hour salt spray test as outlined in Method 509.3, MIL-STD-810. After completion, the test set shall conform to the requirements specified in 3.5.6.

4.9.2 Acceptance examination.

4.9.3 Pre-test information. Before performing any tests, the test set shall be allowed an initial warm up of 5 minutes.

4.9.4 Control circuit tests. With the test set energized as specified in 3.6.1 and the production test cable (see table V) connected to J311 and J312, the numerical indicators shall show proper indication (1 thru 126) on both boards of indicators when tested as specified below.

(a) Position power switch to ON position.

(b) Position the auto-manual switch to manual.

(c) Depress the re-cycle switch to clear circuits.

(d) Position function switch to position No. 3.

(e) Depress start manual advance switch and observe the top group of numerical indicators. The numeral 1 should be showing.

(f) Depress the search switch, and observe the lower group of numerical indicators. The numeral 1 should be showing.

(g) Depress the start manual advance switch. After each depression, the next sequence numerical indicator shall show on each group of numerical indicators. They shall remain "in step" through the test from 1 through 126.

(h) When the indicators have indicated 126 on each group, the next depression of the start manual advance switch will extinguish the top row of the numerical indicators and the "end of test" light shall show.

(i) Depress the recycle switch to clear the circuit and extinguish the bottom row of numerical indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

4.9.5 Voltage test. With the specified voltage in 3.6.1 applied, set the voltage to the calibrate mark on the meter by use of the voltage adjust control and associated push button. Depress the push button to read and adjust. The voltages specified in table VIII shall be measured to ground (chassis).

TABLE VIII. Voltage test.

Test point	Accept voltage	Test condition
28V. output J309 Pin No. All (neg.) on J311 and chassis ground (pos.)	24-32 V. D.C. 500 V.D.C.	110 ohm load function switch on position No. 5. Range switch on R x 1 (red) Depress start manual advance switch to conductor No. 5. TOP numerical indicator to read 5.
Pin No. All (neg.) on J311 and chassis ground (pos.)	1,000 V.D.C.	Function switch on position No. 6. Range switch on R x 1 (red). Depress start manual advance to conductor No. 5. TOP numerical indicator to read 5.

4.9.6 Cycling time. Position the controls as outlined in table IX after connecting the production test cable (see table V) to the test set and energize the set as specified in 3.6.1. Depress the start manual advance switch. The test set shall cycle through all 126 conductors, making a total of 126 indications as noted on the numerical indicators, within the time specified in 3.6.2.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE IX. Control positions during cycling operations.

Control	Position	Condition
Power	ON	Pilot light lit
Auto/manual	Auto	
Re-cycle	Depress	Numerical indicator blank red "No-Go" light ON
Ohmmeter range SW	R x 1 (black)	
Function SW	Position No. 4	Meter deflection to right
Adjustable meter pointer	Full right	
Left zero adj	Depress meter-cal SW and left hand zero	Meter at left end of scale
Right meter adj	Adj for full scale	Full scale deflection
Adjustable meter indicators	Set to 8 ohms	Both indicators at 8 ohms

4.9.7 Single-ended continuity resistance, test function No. 1. With the test set energized as specified in 3.6.1, it shall meet the requirements of 3.6.3.1 when subjected to the following test. Connect production test cable (see table V) to J311 only. Connect shorting plug to other end of production test cable. Position controls as show in table X. Reposition test function switch to position No. 1. Press the start manual advance switch. The test set shall advance to the first conductor, as indicated on the top row of numerical indicators. When the ohmmeter indicator reaches the set acceptance level, the red "No-Go" light shall go off and the green "Go" shall light. Proceed through five steps by operation of the start manual advance switch, observing the same as for the first conductor, except that the numerical indicators shall sequentially progress. Operate the auto-manual switch to auto position. The test set shall step automatically to each successive conductor, rapidly indicating similarly to the manual performance, and progressing to the next conductor when the set acceptance level is indicated on the meter. The test is completed after progressing through 126 conductors, when the "End of Test" light shall be on.

4.9.8 Shorted conductors, test function No. 2. When energized as specified in 3.6.1, the test set shall meet the requirements of paragraph 3.6.3.2 when subjected to the following tests. Connect the production test cable (see table V) to J311 only. Leave the other end of the production test cable open. Position controls as shown in table X. Press start manual advance switch. The test set shall advance to No. 1 conductor as shown on numerical indicators. The green "Go" light shall be lit. Press start manual advance switch through five conductors, observing "Go" light and sequential advance of numerical indicators. Operate the auto/manual switch to "Auto" position. The test set shall now cycle very rapidly through 126 conductors, then show "End of Test" light ON and numerical indicators blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

TABLE X. Control positions and conditions during continuity-resistance test.

Control	Position	Condition
Power switch	ON	Power Indicator ON
Auto/manual switch	Manual	
Re-cycle switch	Operated	Numerical indicators blank
Test function selector switch	Position No. 4	"No-Go" indicator ON
Ohmmeter range switch	R x 1 (black)	Meter indicator at right end of scale
Adjustable meter indicator	Full right	
Left zero adjust	Depress meter cal switch to left hand zero	Meter indicator at left
Right zero adjust	Adjust for full scale	Full scale deflection
Adjustable meter indicator	Set to 8 ohms	Both indicators at 8 ohms.

4.9.9 Transposed conductors, test function No. 3. The test set shall meet the requirements of 3.6.3.3 when subjected to the tests in 4.9.4 and the following: Connect the production test cable (see table V) to J311 and J312. Position controls as shown in table XI, except test function selector shall be on position No. 3. Complete the test as shown in 4.9.8.

4.9.10 Continuity resistance; double-ended, test function No. 4. The test set shall meet the requirements of 3.6.3.4 when subjected to the following tests. Connect the production test cable (see table V) to J311 and J312. Position controls as shown in table X. Proceed as in 4.9.7.

TABLE XI. Control positions and conditions during shorting tests.

Control	Position	Condition
Power switch	ON	Power indicator ON
Auto/manual	Manual	
Re-cycle switch	Operated	Numerical indicator blank
Test function selector	Position No. 2	"No-Go" indicator ON
Ohmmeter range switch	Any	Ohmmeter indicator at left "0"

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

4.9.11 Insulation resistance, 500V potential, test function No. 5. The test set shall meet the requirements of 3.6.3.5 when subjected to the following tests, and to the test in 4.9.12. Connect the production test cable (see table V) to J311 and J312. Position controls as shown in table XII. Proceed as in 4.9.7 except as stated herein.

TABLE XII. Control positions and conditions during insulation resistance tests.

Control	Position	Condition
Power switch	ON	Power indicator ON
Auto/manual switch	Manual	
Re-cycle switch	Operated	Numerical indicator blank
Test function selector switch	Position No. 5	"No-Go" indicator ON
Ohmmeter range switch	R x 1 (red)	Meter indicator at left end of scale
Adjustable meter indicator	Full right	
Voltage adjust switch & control	Depress switch & adjust control to volt adjust	Meter indicator at volt adjust line
Left zero adjust	Adjust to left-hand zero	Meter indicator at left
Right zero adjust	Depress meter Col Switch & adjust to right hand zero	Meter indicator full-scale deflection to right
Adjustable meter indicator	Set to 10 megohms	Pointer set to 10 megohms

4.9.12 Insulation resistance, 1000 V potential, test function No. 6. Connect the production test cable (see table V) to J311 and J312. Position controls as shown in table VIII, set ohmmeter range switch in position R x 100 meg. (red) and position function selector on position 6. Proceed as in 4.9.7 except as stated herein.

4.9.13 Ohmmeter, test function No. 7. The test set shall meet the requirements of 3.6.3.7 when subjected to the following tests. With the test set energized as specified in 3.6.1, place the function switch on position No. 7. Connect ohmmeter test leads to "Ext. ohms" jacks. Position ohmmeter range switch on R x 1 (black). Set left zero adjustment for left zero on scale. Bring ohmmeter test leads together and adjust for right scale zero. Recheck left scale, separating leads. Repeat on R x 10, R x 100, R x 1K, R x 10K, R x 100K and R x 1 meg scales. Each scale extreme shall be adjustable with its

Source: https://assist.dla.mil -- Downloaded: 20260-22T22:38Z
Check the source to verify that this is the current version before use.

associated control, with not less than 30° of rotational adjustment still available.

4.9.14 Attenuation, R. F. test function No. 8. The requirements of 3.6.3.8 shall be met when the test set is subjected to the following tests. With the test set energized as specified in 3.6.1, position all R. F. attenuator switches down, or at the "IN" position. P4 and P11 shall be connected to their respective sockets. Place the function switch on position No. 8. Allow the R. F. attenuation test module to warm up for at least 5 minutes to reach stability. Rotate "Zero adj" control. It shall be possible to drive the zero indicator from + 1 db through 0 to full scale with not less than 45° rotation and not more than 135° rotation of "Zero adj" control. Rotate "Zero adj" control so that the zero indicator reads 1 db. Operate the 1 db attenuation switch to the up position (out). The zero indicator shall now read "0" ± .15 db/10 db.

4.9.15 Accuracy test.

4.9.15.1 Setup. The accuracy requirements of 3.6.3.1 shall be met when the test set is subjected to the following tests. Position the controls as in table X, except that the adjustable meter indicator shall be at full right. Connect the production test cable (see table V) to J311 only. Using clip leads, connect the decade resistor box (see table V) to all connectors and K6 of the free end of the production test cable. Operate the start manual advance switch until conductor No. 5 shows on the numerical indicators. Turn the test functional selector switch to position No. 1. Using the 10 ohms setting in the decade resistor box, the ohmmeter should now indicate 10 ohms ± 10 percent. Operating the decade box will permit readings to be taken at 1 to 10 ohms in 1 ohm steps, 10 to 100 ohms in 10 ohm steps, and 100 to 1,000 ohms in 100 ohm steps. The meter shall indicate these values, ± 10 percent.

4.9.15.1.1 Operate ohmmeter range switch to R x 10. Set the decade box (see table V) for zero ohms. Re-zero both ends of the ohmmeter scale as shown in table X. Set the decade box for 100 ohms. The ohmmeter shall indicate this value (10 ohms on R x 10 range) ± 10 percent (see 3.6.3.7b).

4.9.15.1.2 Operate ohmmeter range switch to R x 100. Set the decade box for zero ohms. Re-zero both ends of the ohmmeter scale as shown in table X. Set the decade box for 1,000 ohms. The ohmmeter shall indicate this value (10 ohms on R x 100 range) ± 10 percent (see 3.6.3.7c).

4.9.15.1.3 Operate ohmmeter range switch to R x 1K. Set the decade box (see table V) for zero ohms. Re-zero both ends of the ohmmeter scale as shown in table X. Set the decade box for 10 K ohms. The ohmmeter shall indicate this value (10 ohms on R x 1K range) ± 10 percent (see 3.6.3.7d).

4.9.15.1.4 Operate ohmmeter range switch to R x 10K. Set the decade box (see table V) for zero ohms. Re-zero both ends of the ohmmeter scale as

Source: https://assist.dla.mil -- Downloaded: 20260-22T22:38Z
Check the source to verify that this is the current version before use.

shown in table X. Set the decade box for 100K ohms. The ohmmeter shall indicate this value (10 ohms on R x 10K range) ± 10 percent (see 3.6.3.7e).

4.9.15.1.5 Operate ohmmeter range with switch R x 100K. Set the decade box (see table V) for zero ohms. Re-zero both ends of ohmmeter scale as shown in table X. Set the decade box for 1 megohm. The ohmmeter shall indicate this value (10 ohms on R x 100K range) ± 10 percent (see 3.6.3.7f).

4.9.15.1.6 Operate ohmmeter range switch to R x 1 meg. Set the decade box (see table V) for zero ohms. Re-zero both ends of the ohmmeter scale as shown in table X. Set the decade box for 10 megohms. The ohmmeter shall indicate this value (10 ohms on R x 1 meg range) ± 10 percent (see 3.6.3.7g).

4.9.15.2 Accuracy requirements. The accuracy requirements of 3.6.3.5 shall be met when the test set is subjected to the following tests. Position the controls as in table XII except that function selector switch shall be on position No. 6, and the ohmmeter range switch shall be on R x 1 (red). Set the adjustable meter indicator to full scale right. Connect the decade resistance box (item 2) as stated in 4.9.15.1. Operate the decade box to present 2 megohms. Operate the start manual advance switch until conductor No. 5 shows on the numerical indicators. The ohmmeter shall indicate 2 megohms (2 megohms on R x 1 scale) ± 10 percent.

4.9.15.2.1 Test 1. Operate re-cycle switch. Reposition the ohmmeter range switch to R x 10 (red). Re-zero both ends of the ohmmeter scale as shown in table XII. Set the decade box (see table V) for 20 megohms. Operate the start manual advance switch until the No. 5 conductor shows on the numerical indicators. The ohmmeter shall indicate 20 Megohms (2 megohms on R x 10 scale) ± 10 percent.

4.9.15.2.2 Test 2. Operate re-cycle switch. Reposition the ohmmeter range switch to R x 100 (red). Re-zero both ends of the ohmmeter scale as shown in table XII. Set the decade box (see table V) for 200 megohms. Operate the start manual advance switch until the No. 5 conductor shows on the numerical indicators. The ohmmeter shall indicate 200 Megohms (2 megohms on R x 100 scale) ± 10 percent.

4.9.15.3 R. F. attenuator accuracy requirements. The accuracy requirements of 3.6.3.8 shall be met when the R. F. attenuator test module is subjected to the following test. Attach UG connectors of P-4 and P-11 to the external step attenuator (see table V). Turn the test function selector switch to position No. 8 and allow 5 minutes warm-up. With the R. F. meter adjusted for zero, compare the accuracy of each attenuator module step with the similar db value of the external unit (see table V). Each step shall measure within the required accuracy of .15 db/10 db.

4.9.16 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

requirements of MIL-STD-1190, packaging data sheet for APN 8213077, and specifications referenced therein.

5. PACKAGING

5.1 Preservation, packing, unitization and marking. Level A, B or C as specified (see 6.2).

5.1.1 Level A or B. Preservation, packing, unitization, and marking shall be in accordance with packaging data sheet APN 8213077 and specifications referenced therein.

5.1.2 Level C. Level C preservation, packing, unitization and marking shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers a test set capable of testing cable assemblies for conductor resistance, insulation resistance continuity, crossed conductors, shorted conductors, and attenuation of coaxial conductors. The test set is capable of testing cables having 126 conductors or less, and/or assemblies having six coaxial conductors or less. The test set is portable and will operate at either 115 or 230 V.A.C. at 55 to 410 Hz and is expected to be used mainly in the field. Either one end or both ends of the cable may be connected for test.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. When first article is required (see 3.1 and 4.4)

d. First article sample size including date of delivery (see 4.4)

e. Quality conformance inspection requirements (see 4.5)

f. Applicable stock number

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the currentversion before use.

g. Lot formation (see 4.5.1)

h. Inspection of components requirements (see 4.6)

i. Required inspection equipment (see 4.7)

j. Required preservation, packing, unitization and marking (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Accessories. Coupling media are required to adapt the various connectors used on cables to the test set.

6.5 Definitions.

6.5.1 Standard measuring equipment (SME). Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.5.2 Tactile. Skillful manipulation or sensing; of, having, or related to the sense of touch.

6.6 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.7 Subject term (keyword) listing.

Coaxial conductor testing
Decade box
Evaluation equipment
Resistance testing

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-MI

Preparing Activity:
Army-MI

Review activity:
Army-AR

Project No. 6625-A988

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-55207C (MI)	2. DOCUMENT DATE (YYMMDD) 1 April 1991

3. DOCUMENT TITLE
Test Set, Electrical, Cable: AN/GSM-45

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME Commander U.S. Army Missile Command	b. TELEPHONE (Include Area Code) (1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST Redstone Arsenal, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:38Z
Check the source to verify that this is the current version before use.